CRM Customer Segmentation - VarGroup Project

Group 1 - Master in Data Science and Business Analytics

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Business Context & Challenge

Moving from a single database "Hidden" Clusters.

Mass Marketing" causing inefficiency and high Churn

Understanding the Italian Consumer

The Challenge

Market Context

The Conflict

The Pivot

Strategic Insight

Loyalty = Profitability

Data-Driven Precision

Saturated retail market

Objectives: Defining Customer Value

Goal

Our goal was to segment the customer base into 4 actionable profiles to address three core business needs

Retention

How do we use discounts to win back the 'At-Risk' customers who haven't visited in over 90 days?

Loyalty

How can we reward our VIPs through the app and keep our Traditional shoppers coming back to the store?

Development

Develop the potential of 'Recent Regulars' by using personalized cross-selling to increase their average basket value and transition them into higher-value tiers.

The RFM Framework & Business Logic

To solve the challenge, we adopted the RFM Model, focusing on:

df_receipts_aggregated.csv

CATEGORY_DESC	QUANTITY	SALES	qty_min	qty_max	qty_mean	qty_std	sales_min	sales_max	sales_mean	sales_std
ACCESSORIES	102355.0	371479.97	1.0	1000.0	1.678694	9.415763	0.00	199.75	6.092532	11.336296
BEEF_PORK	33105.0	366165.40	1.0	7.0	1.071498	0.295133	5.50	74.70	11.851547	3.954431
DRINKS	69033.0	407209.88	0.0	72.0	2.017801	2.438991	0.00	370.45	11.902545	14.415995
FISH	298738.0	3473112.20	1.0	24.0	1.083959	0.366487	0.00	275.00	12.602049	5.948863
FRIES	169226.0	793951.00	1.0	21.0	1.116001	0.481350	3.30	135.45	5.235900	2.781752
FRUIT	99150.0	721835.95	1.0	36.0	1.136807	0.520773	0.00	118.80	8.276227	5.248269
GADGETS	6020.0	0.00	1.0	62.0	1.050794	1.352903	0.00	0.00	0.000000	0.000000
OTHER	31932.0	1886.50	1.0	24.0	1.215670	0.584100	0.00	53.90	0.071820	1.485593
OTHER_FROZEN	354583.0	2655466.43	1.0	40.0	1.077907	0.406419	0.00	318.00	8.072431	3.394915
POULTRY	118267.0	1340303.80	1.0	10.0	1.078980	0.320844	0.00	139.50	12.227934	4.452578

This logic allowed us to transform **raw receipts** into 4 distinct "Personas", making the results directly usable by the Marketing Department.

Exploratory Data Analysis

Customers

25727 Customers

Sales

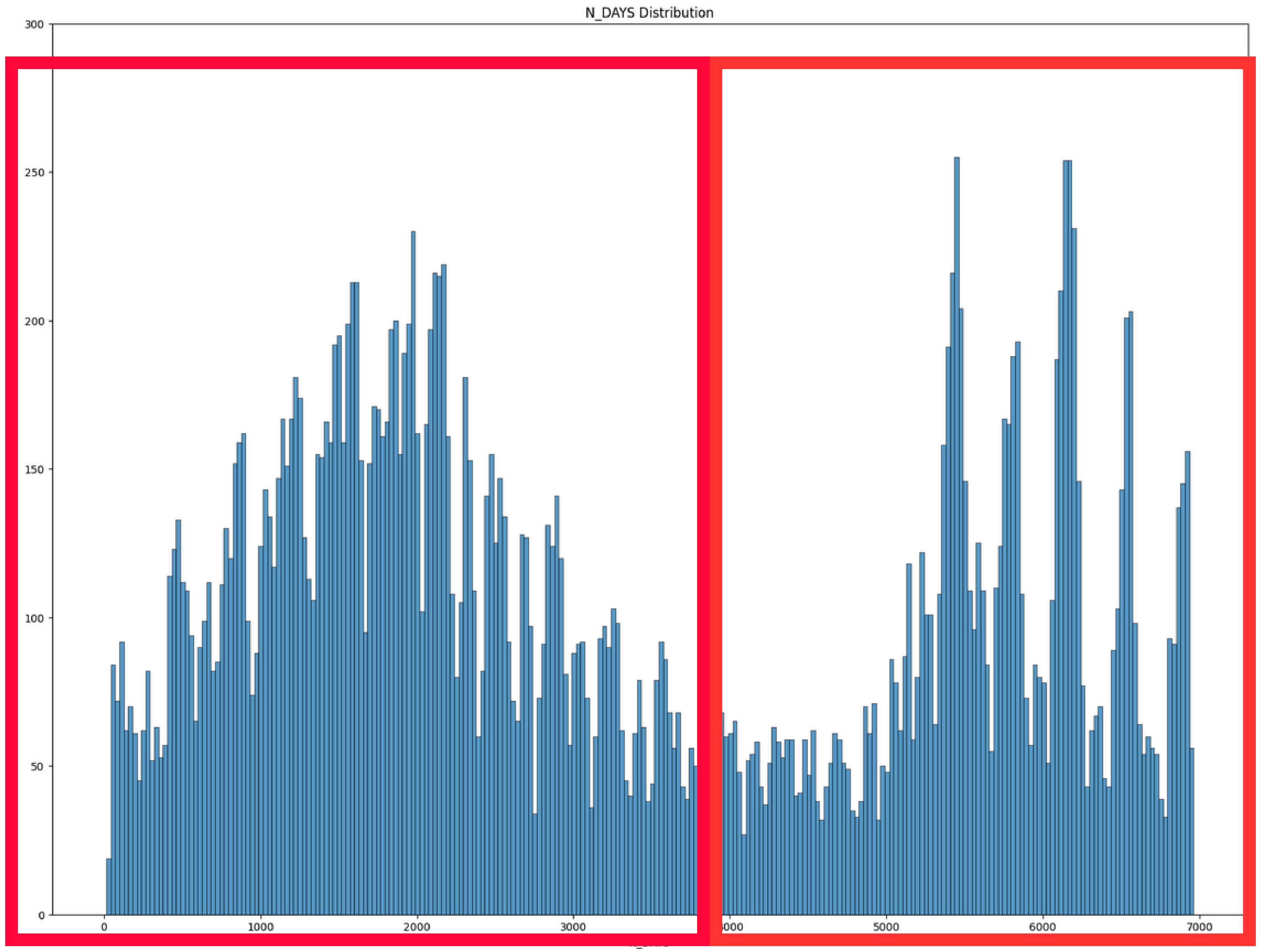
16,521,041.81 euros in Sales

Payment Method

642.167 euros Sales/person

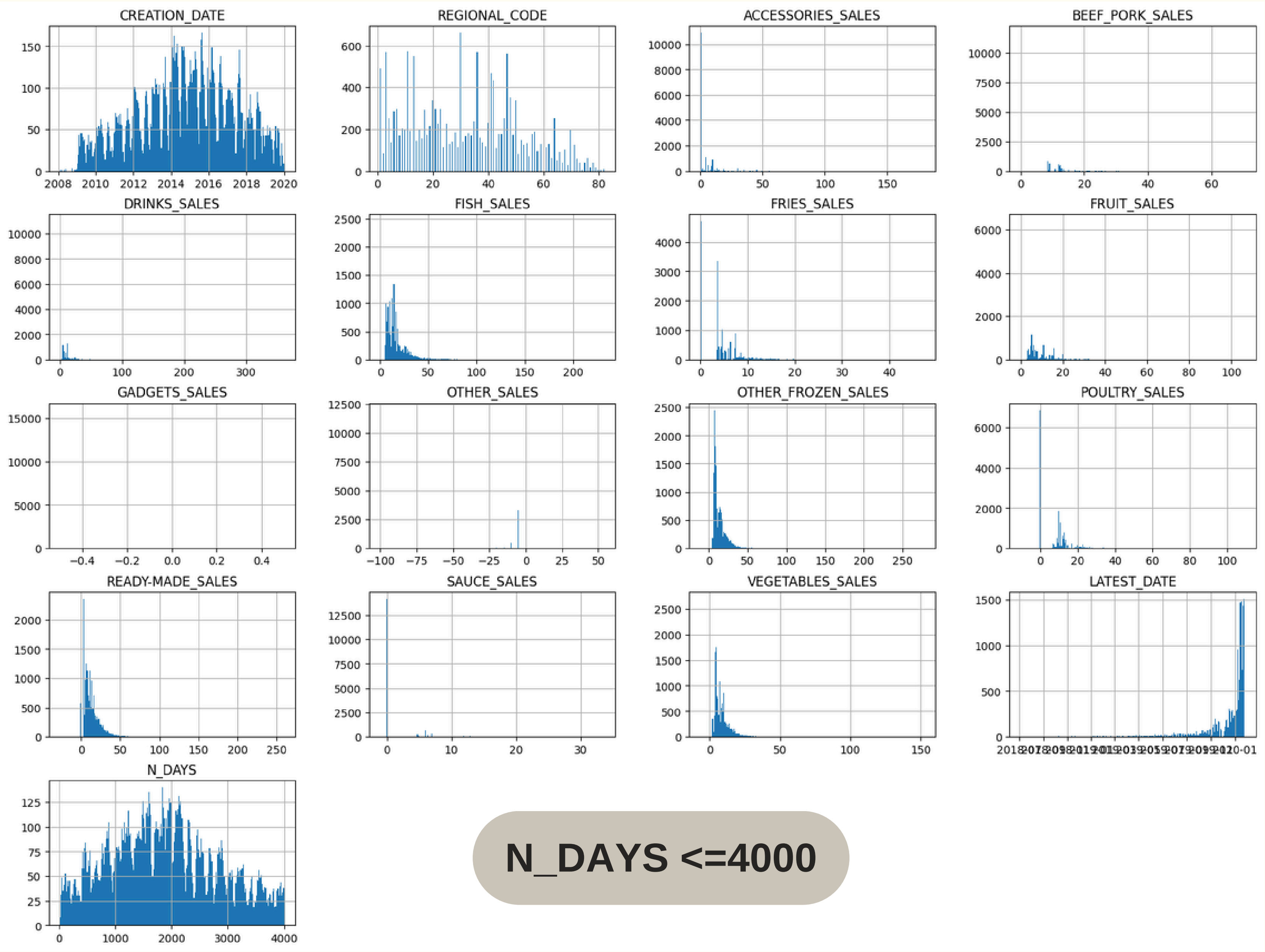
EDA: Customers

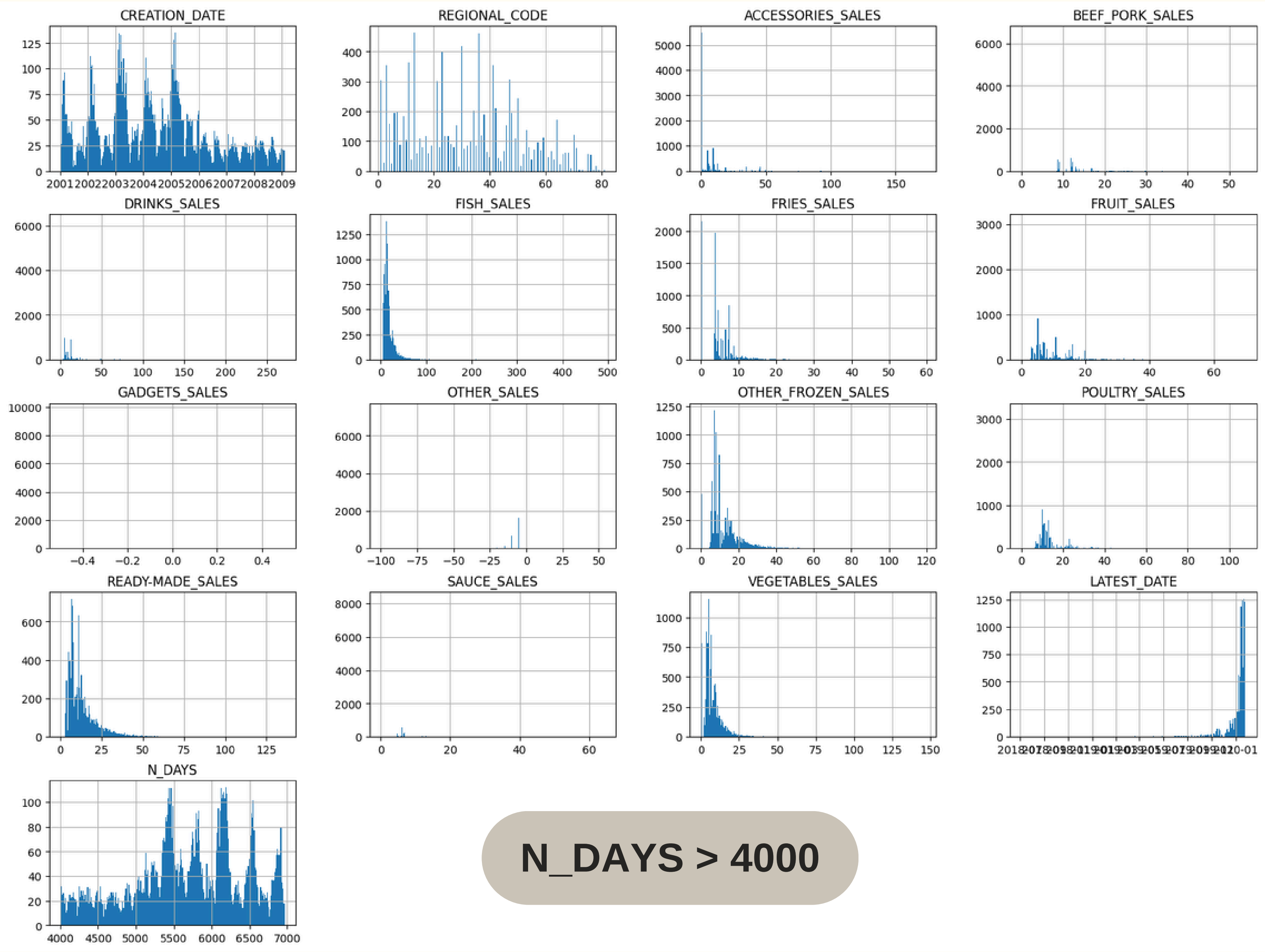
N_DAYS represents the difference of the latest and the creation date
N_DAYS = LATEST_TRANSACTION_DATE - CREATION_DATE



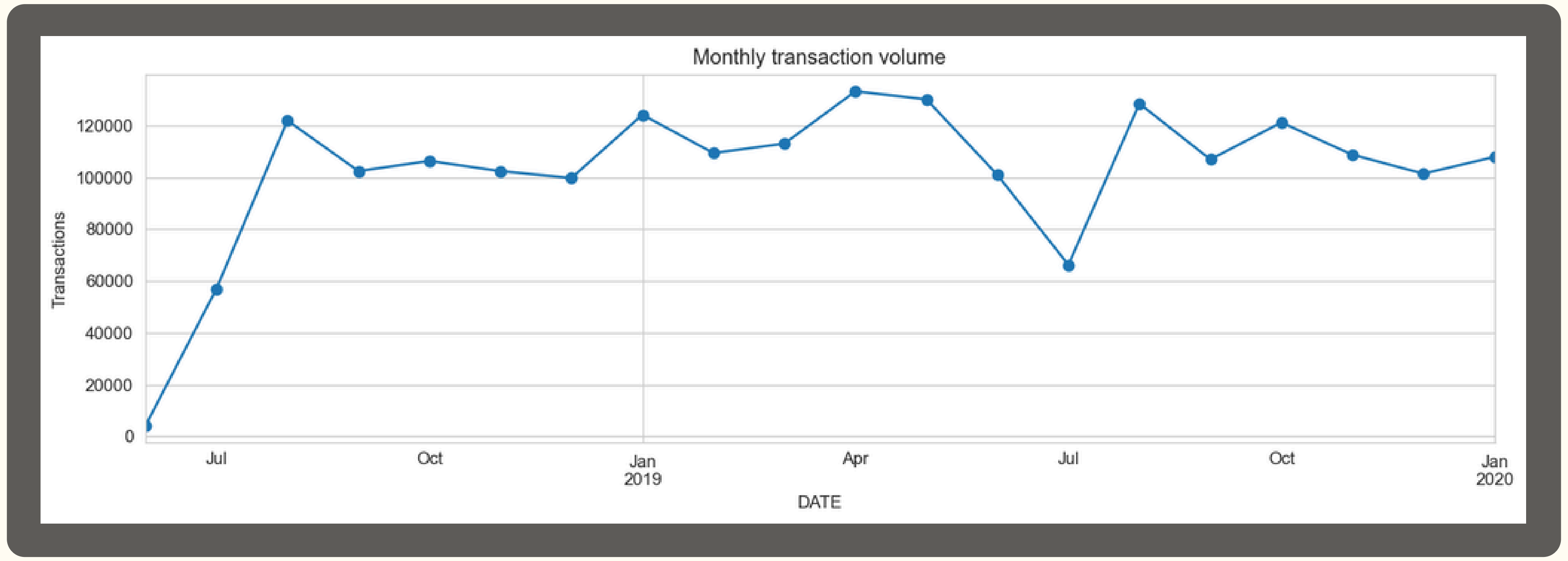
It likely has 2 distributions (bimodal distribution)

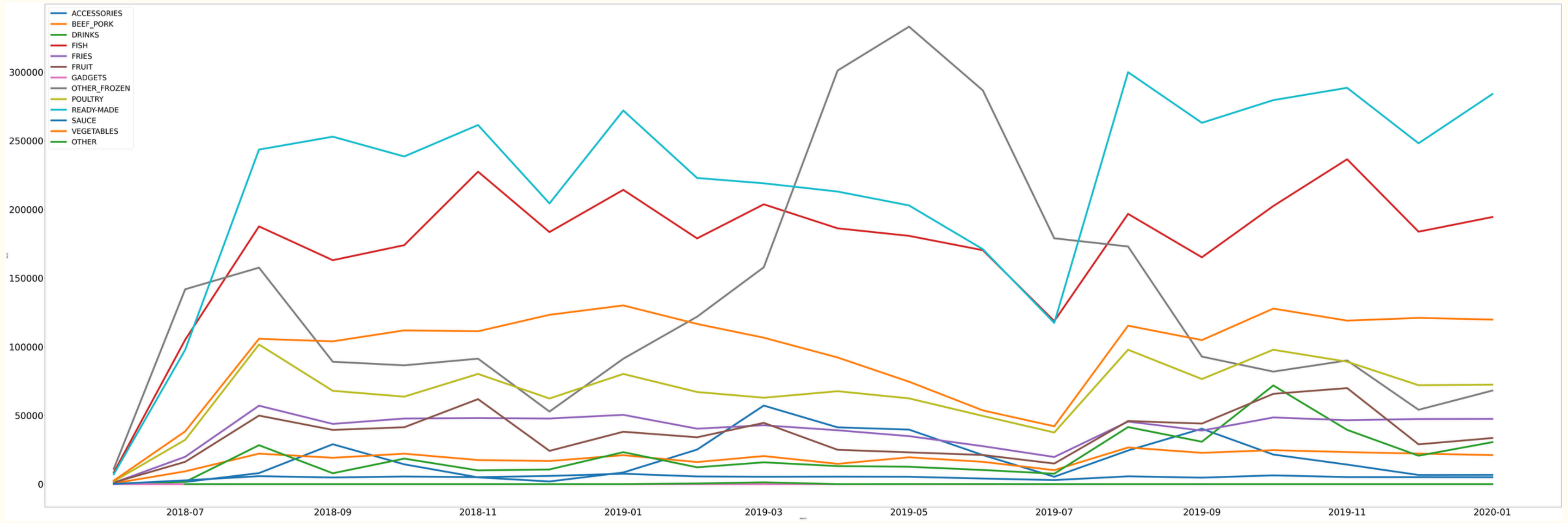
EDA: Customers





EDA: SALES





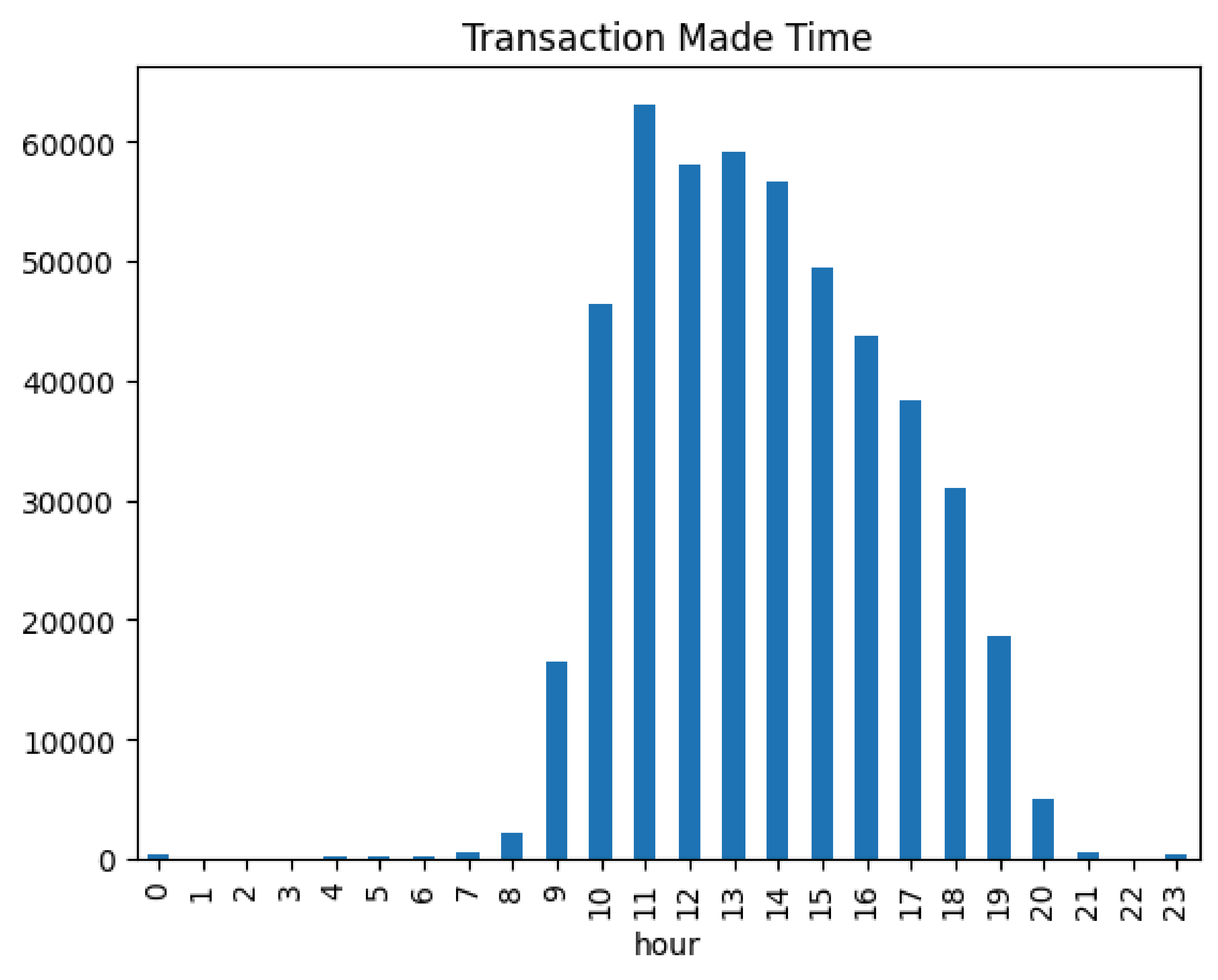
The data doesn't show a clear upward or downward trend.

COVID-19 Pandemic

EDA: SALES

The Morning Ramp-Up

Sales are almost non-existent before 8:00 AM.



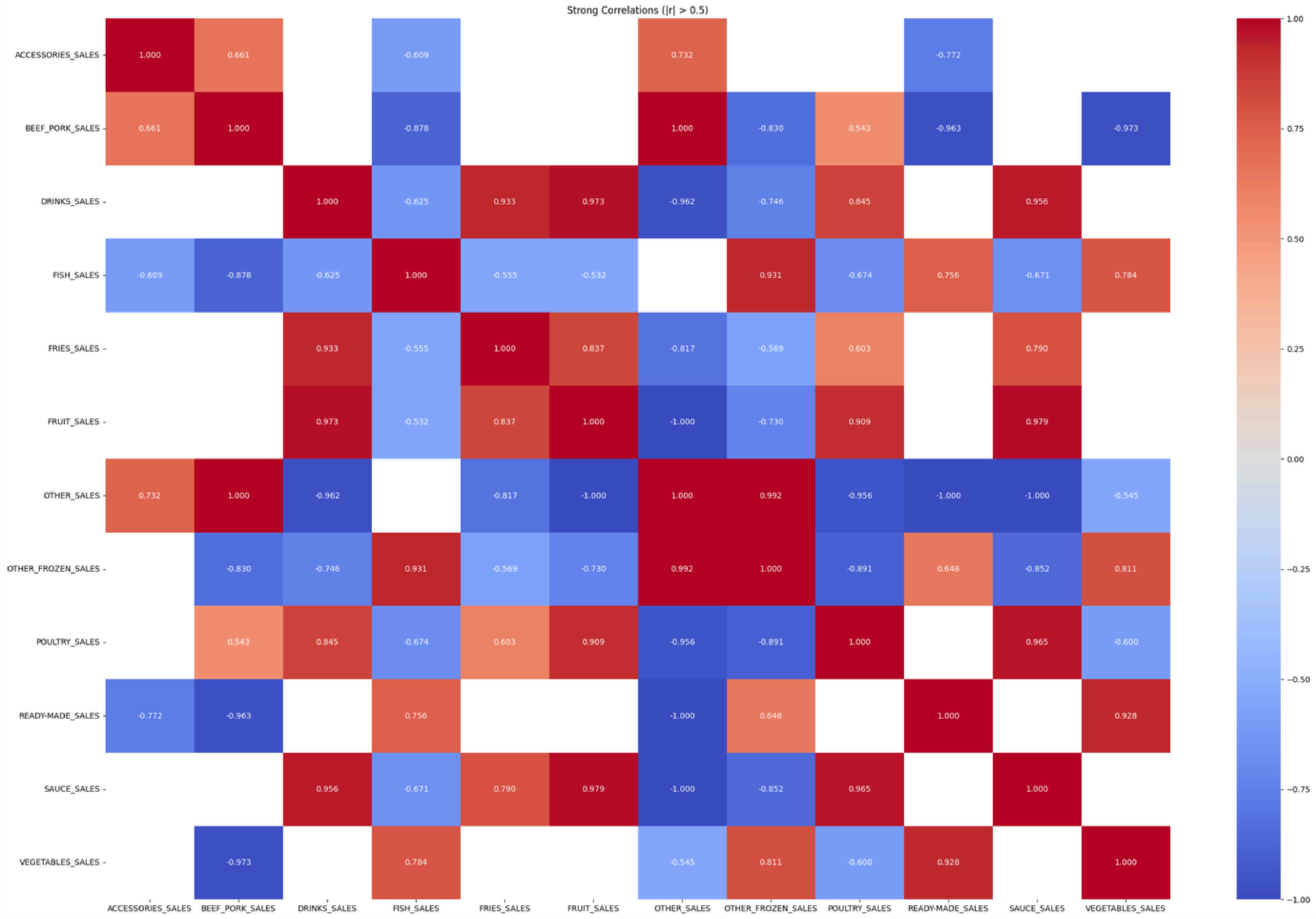
Peak Sales Hours

Hour 11 (11:00 AM) is the busiest time of the day, with volume exceeding 60,000 transactions

Dead Hours

From 11:00 PM to 7:00 AM, the business sees nearly zero transactions.

EDA: PAYMENT METHOD



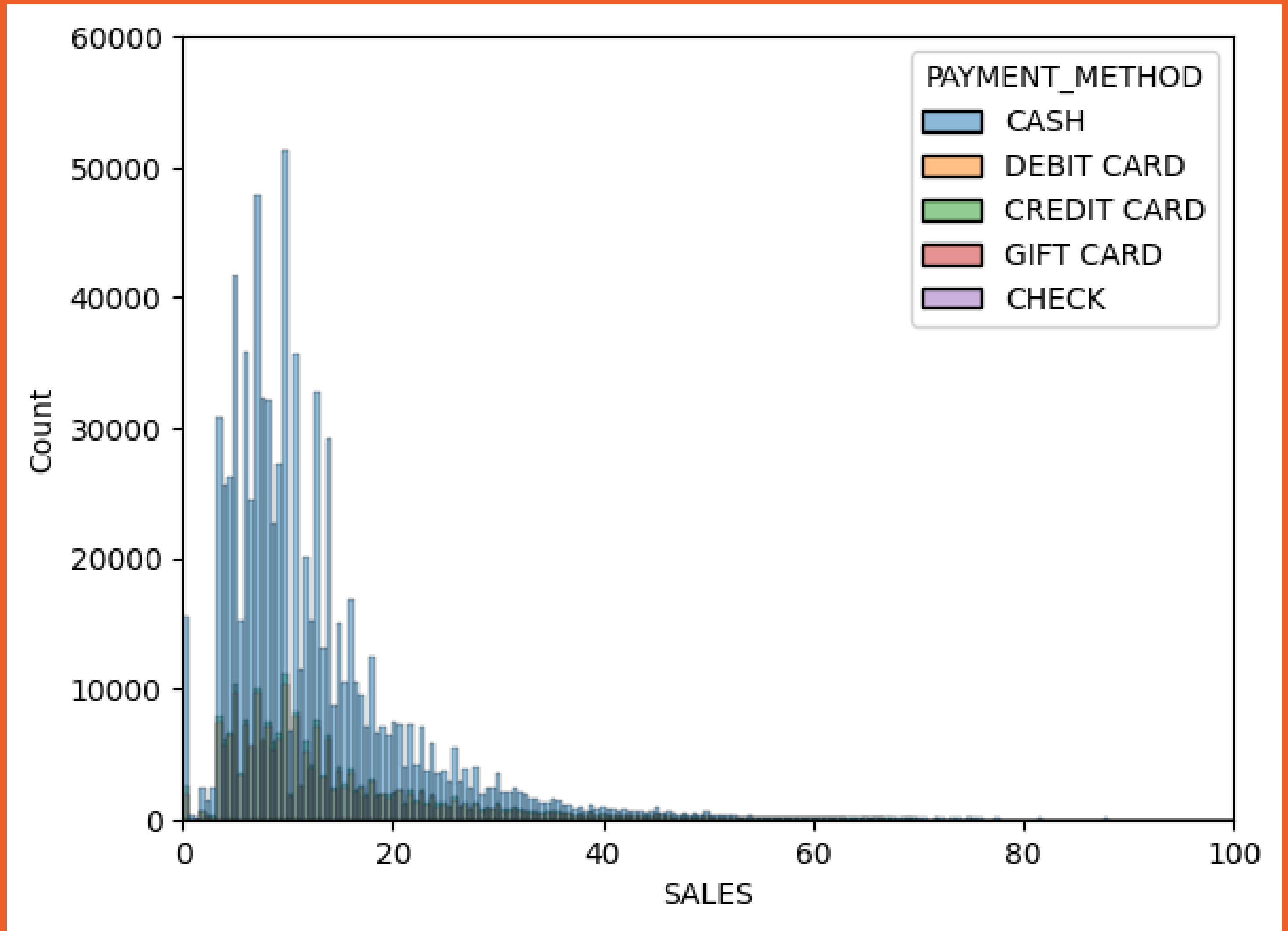
Strong Positive Correlations:

1. **Sauce and Fruit (0.98)**
2. **Fruit and Drinks (0.97)**
3. **Poultry and Sauce (0.96)**

Strong Negative Correlations:

1. **Beef/Pork and Vegetables (-0.97)**
2. **Beef/Pork and Ready-Made (-0.96):**

EDA: PAYMENT METHOD

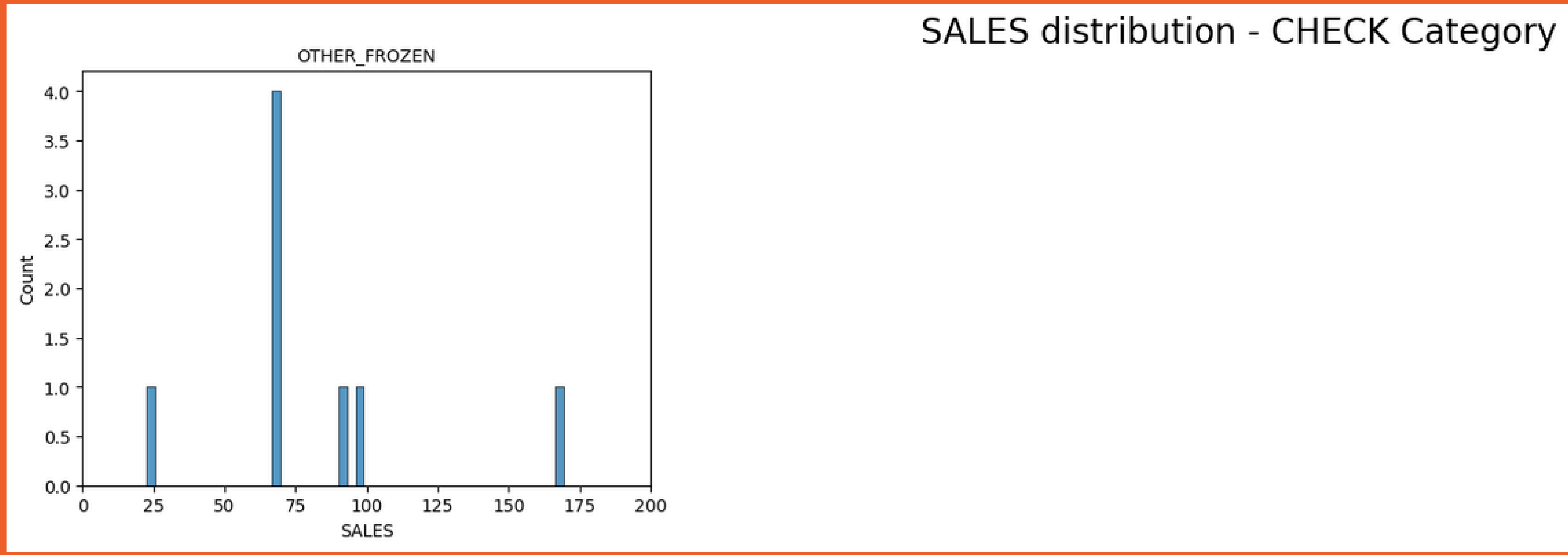


Dominance of Cash

Low Volume Distribution

Sample Biased

With Check, 100 percent buying other frozen category



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Feature Engineering

A small definition

- **Feature engineering** is the process of transforming raw data into relevant features at customer-level. Algorithms can't group receipts, they group *people*.
- Selecting and creating features to help ML algorithms make accurate predictions. The goal: create a "**Customer DNA**".

The RFM Framework

- We selected the industry standard, the **'RFM' Framework** (Recency, Frequency, and Monetary) - We wanted to understand who shopped yesterday, who visits every week, and who spends the most.
- Nevertheless, for a modern CRM strategy, RFM alone is **not enough**.

Top five key features

- We needed deeper **psychological** and **behavioral** insights and looked for several **custom metrics**, building them from scratch.
- A **top five** of these features: Discount Sensitivity, Card Usage Rate, Loyalty Dynamics, Customer Tenure, Category Diversity

Feature Engineering

Feature	Source	Why it matters
RECENCY	Last purchase date	Recent buyers are more engaged
FREQUENCY	Distinct shopping days	Measures loyalty and habit
MONETARY	Gross spend (positive prices only)	True purchase value
AVG_BASKET_GROSS	Mean spend per item	Distinguishes premium vs budget shoppers
CATEGORY_DIVERSITY	Distinct categories bought	Wide vs narrow shopping behaviour
OUTLET_DIVERSITY	Distinct stores visited	Loyal to one store vs multi-store shopper
CARD_USAGE_RATE	Fraction of card payments	Proxy for digitisation and demographics
SPECIAL_RATE	Fraction of promo items	Promo sensitivity
AVG_PTS_EARNED	Mean points on earning receipts	Loyalty programme engagement
REDEMPTION_FREQ	Fraction of receipts with redemption	Active redeemers vs passive earners
DISCOUNT_SENSITIVITY	Discount spend / gross spend	How much the customer relies on coupons
RETURN_RATE	Returns / total transactions	Dissatisfaction or bulk-buy behaviour
TENURE_DAYS	Days since registration	Long-term vs new customers

Feature Engineering - Top five features

Discount Sensitivity

We calculated the exact **ratio of customers' discounted spend** versus their **gross spend** to **isolate** those customers who only walk into the store when there's a heavy promotion.

$$1 - \frac{\text{Net Spending}}{\text{Monetary}}$$

Card Usage Rate

We mapped **payment methods** to understand digitization. By understanding whether payments were made via **card** or **cash**, we created an **indicator** to identify **digital** versus **physical** users.

Loyalty Engagement

We moved beyond just looking at **'Points Earned**', the **'Redemption Frequency'** was crucial as well. This establishes a fundamental distinction by separating **'Passive Earners'** (who just collect points) from **'Active Fans'** (who actually engage with the rewards).

Customer Tenure

A **breakthrough** for our clustering. By calculating the Days since Registration, we could finally distinguish between our '**New Regulars**' and our '**Historical**' customers. We'll see in the results that Tenure was crucial in identifying our long-term customer base.

Category Diversity

We tracked how many different **departments** a customer visits. This feature tells us if they just stop by for a **single item** or if they rely on the retailer for their entire **weekly grocery shop**.

Preprocessing for Clustering

Feature selection

MONETARY is excluded from clustering.

Outliers

Isolating extreme **outliers**.

Capping + Log Transformation

Capping at the 99th percentile and log transformation.

Scaling

MinMax scaling.

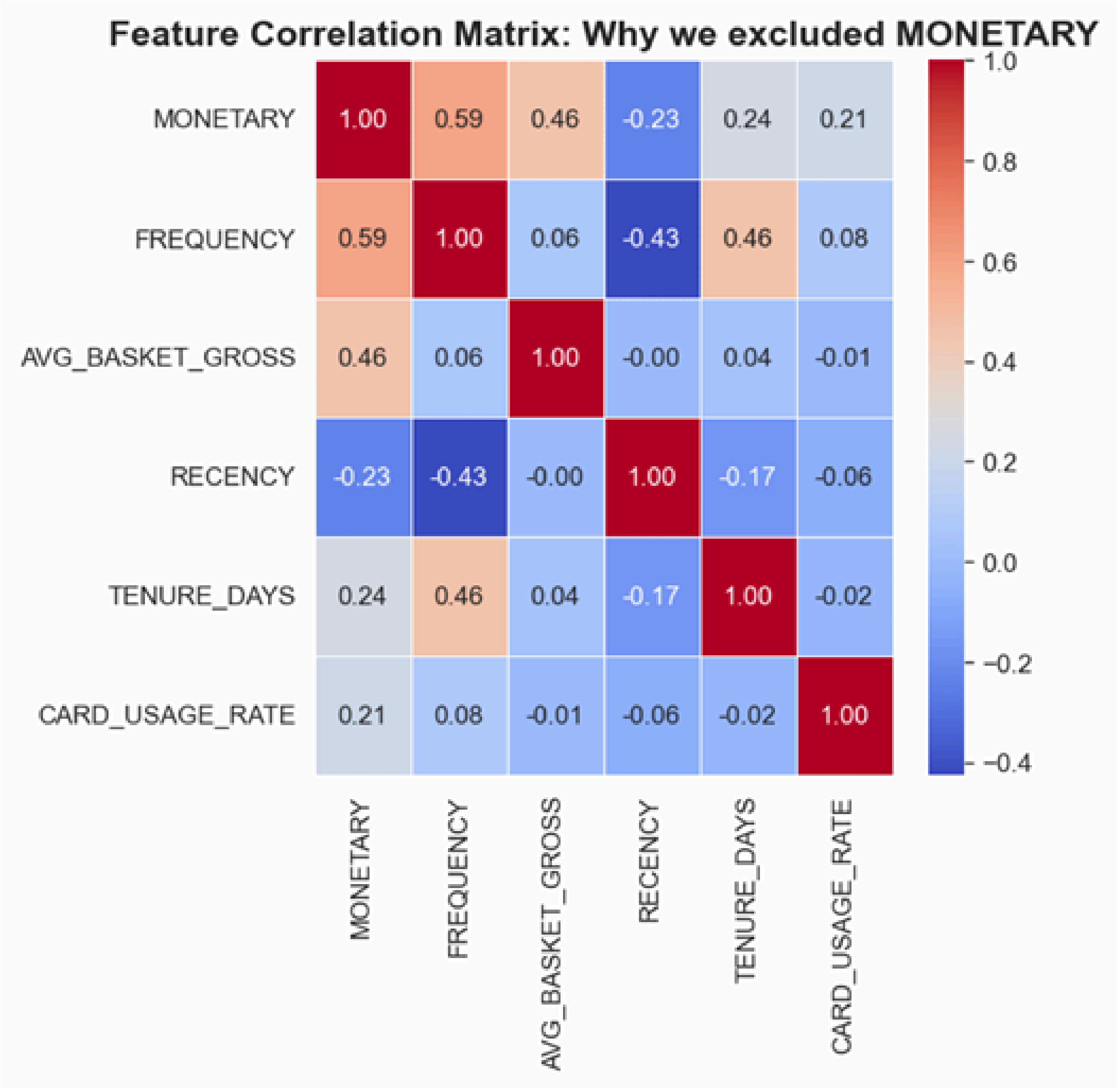
01 Feature selection

02 Outliers

03 Capping

04 Scaling

Letting the Model Discover Real Behavior, Not Just Spending



First, we removed the MONETARY variable from the model because it is strongly correlated with frequency and average basket.

0.59 with FREQUENCY
0.46 with AVG_BASKET_GROSS

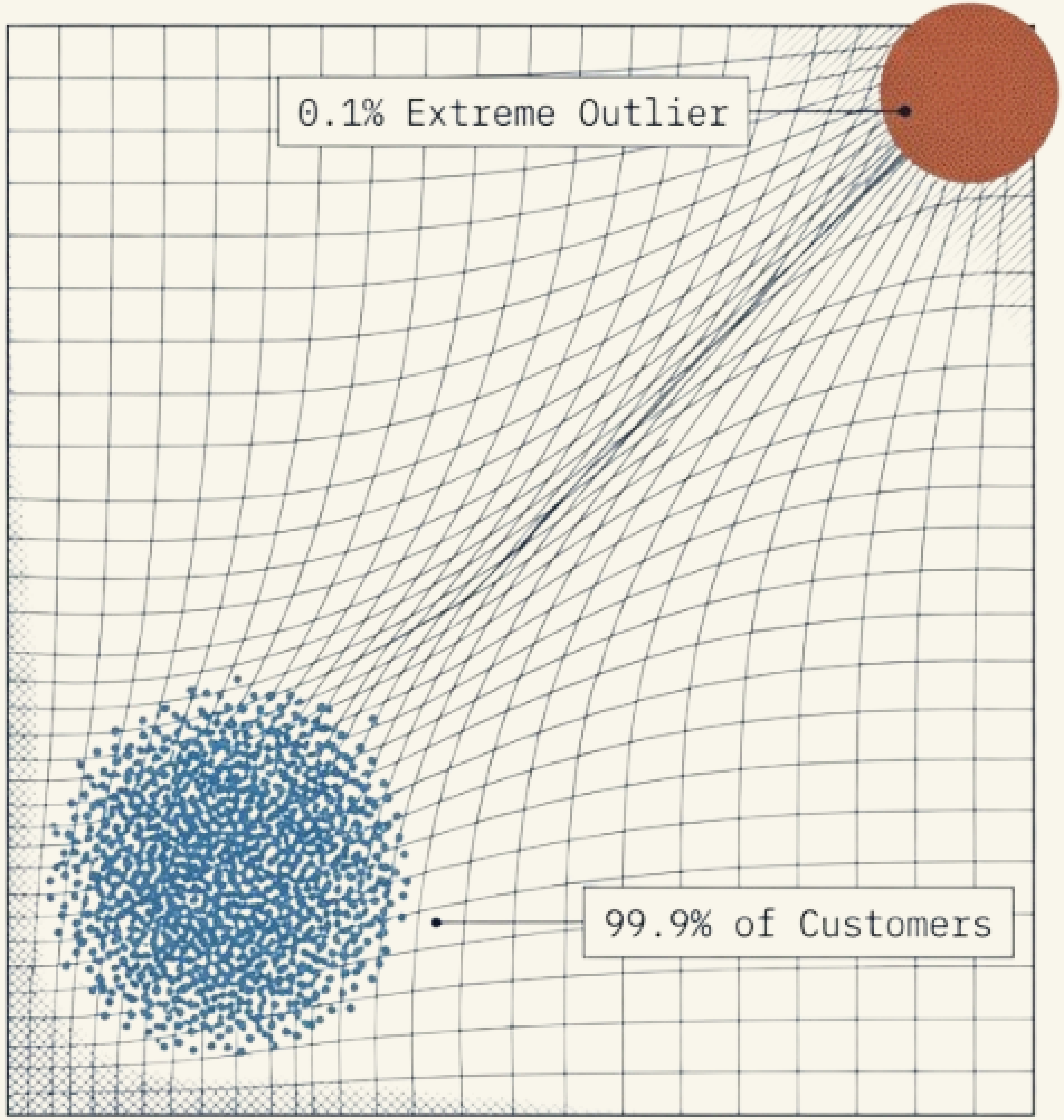
Basically, total spending already comes from how often customers buy and how much they spend each time.

The model would count the same information multiple times and focus only on rich customers.

Protecting the Clusters from Distortion (outliers)

We separated extreme customers from normal ones, about 0.1% of the data. These are customers with very unusual behavior, like extremely high spending or returns.

If we keep them, they distort the clusters for everyone else.



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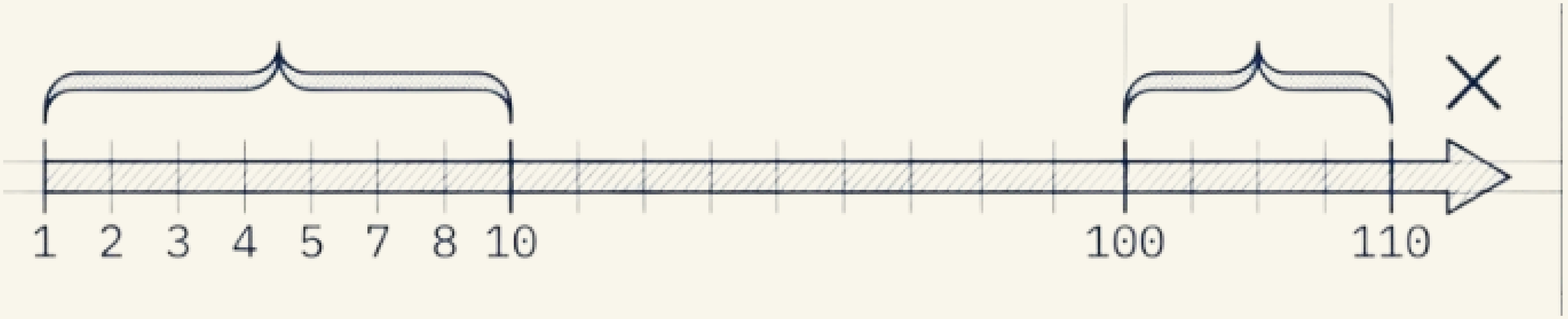
Reducing the Gap Between Small and Very Large Customers

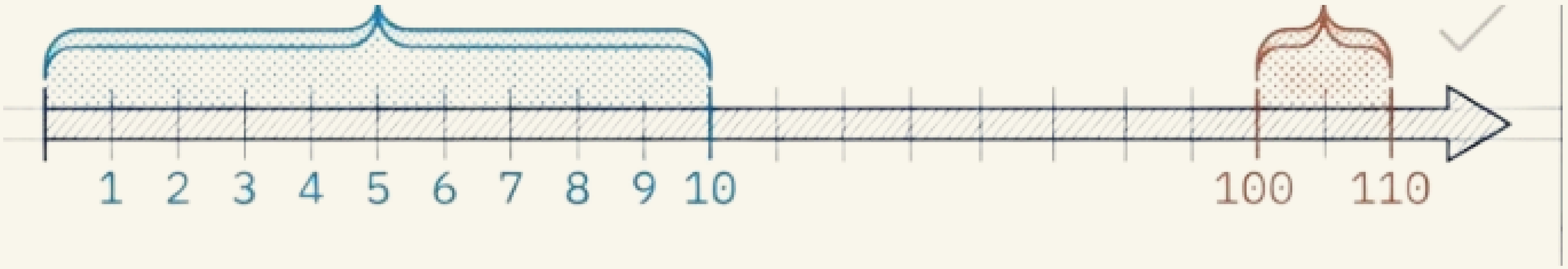
Next, we reduced the impact of very large values. We capped extreme values and applied logarithm.

For example, the difference between 1 and 10 visits is important, but between 100 and 110 is not.

Linear Space (Mathematical Distance)



Logarithmic Space (Behavioral Impact)



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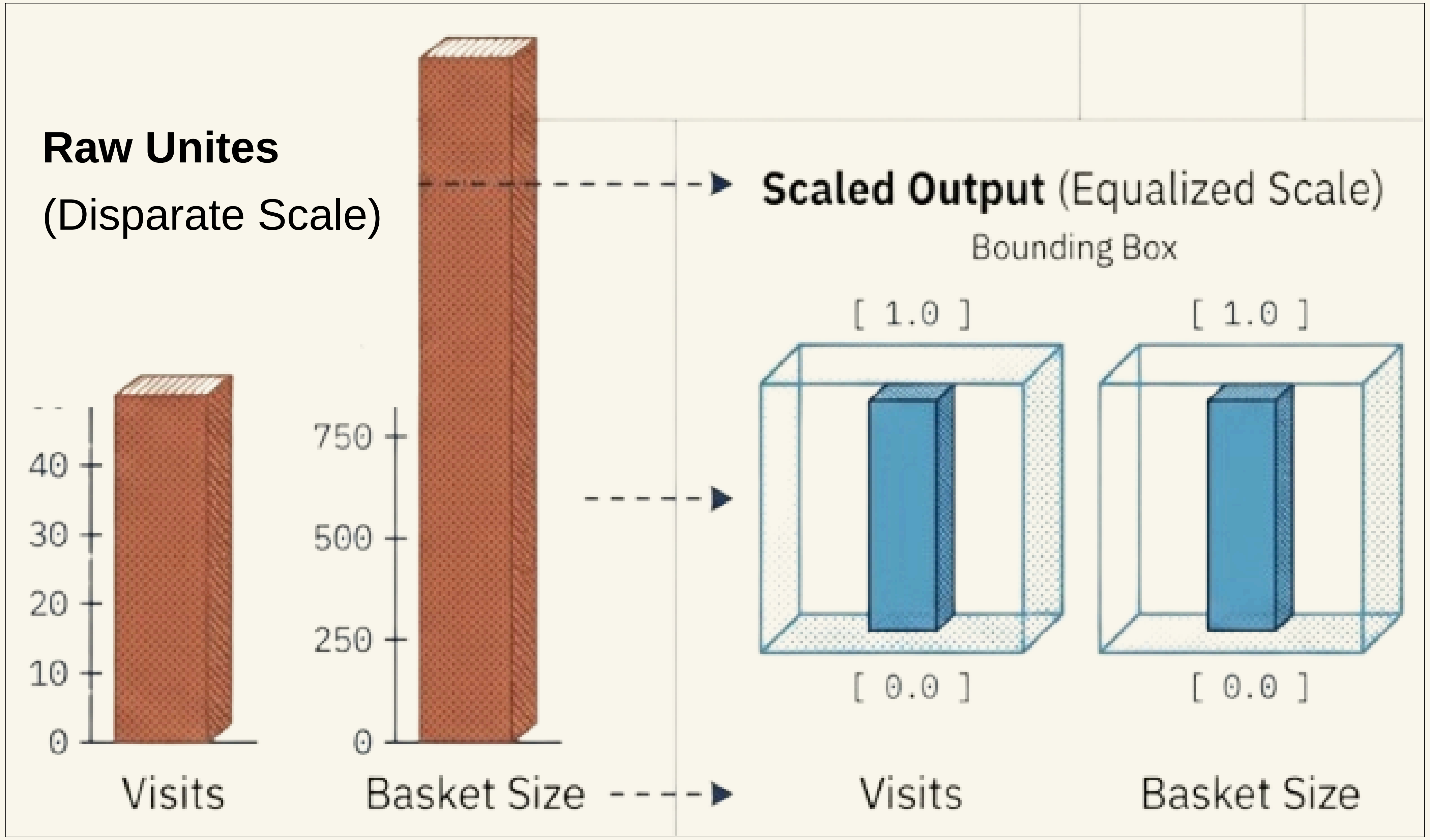
Ensuring All Variables Have the Same Importance

Method of scaling

MinMax

Finally, we scaled all features between 0 and 1

This ensures that all variables have the same importance in the distance calculation.



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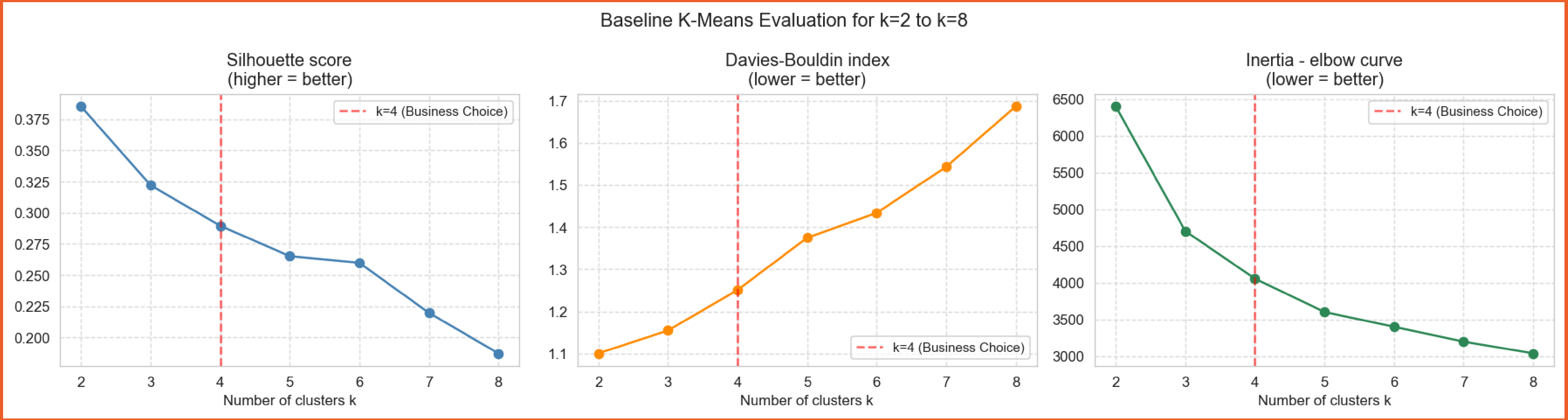
Choosing the Number of Clusters k



silhouette score = 0.2896

DB index = 1.2505

Inertia = 4,057

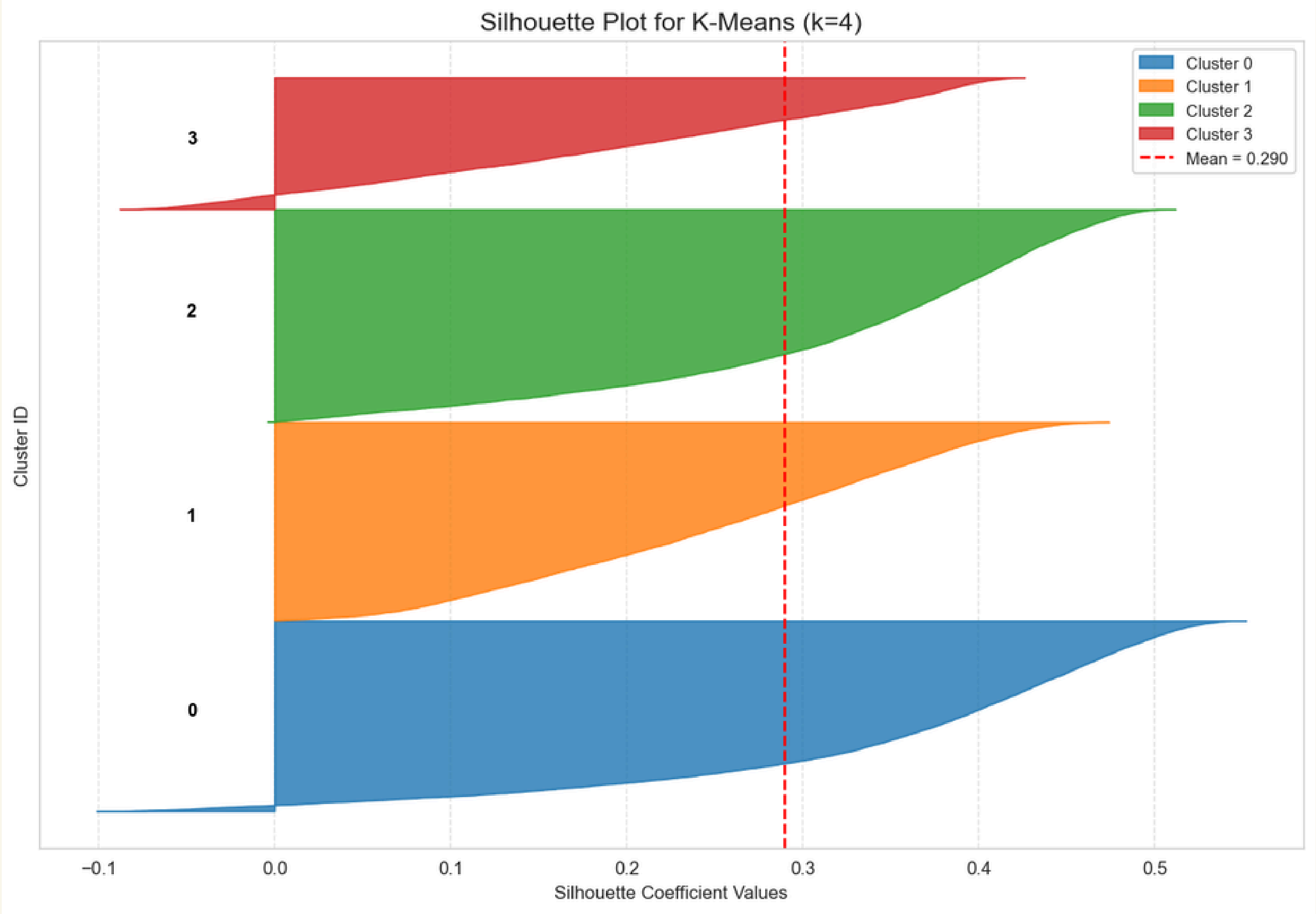
Mathematically, k=2 would be optimal

→

The business decision: k=4

Model : K-Means

A STABLE AND BALANCED CUSTOMER ECOSYSTEM



The distribution is balanced: no single giant cluster absorbing everything, and no near-empty cluster a positive indicator of a robust model.

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Alternative Model Testing

K-Means k=4

- Cluster 0 → 6,684 (26%)
- Cluster 1 → 6,959 (27%)
- Cluster 2 → 7,451 (29%)
- Cluster 3 → 4,634 (18%)

✓

GMM

- Cluster 0 → 5,117
- Cluster 1 → 1,459
- Cluster 2 → 10,176 (~40%) ⚠
- Cluster 3 → 8,926

DBSCAN

- Only 1 cluster : 25,078 (97.7%)

⚠

CRISP DM: Evaluation & Profiling

Objective

Maximize business actionability over abstract mathematical perfection.

The PCA Decision

Excluded Dimensionality Reduction (PCA) to maintain 100% transparency of raw CRM features (Euros, Days, Percentages).

The k=4 Selection

Sacrificed marginal silhouette score (~0.289) to ensure four distinct, manageable marketing personas.

Robustness

Extreme outliers (top 0.1%) isolated during training and mathematically projected back into centroids.

Cluster Centroids Analysis n=25,678 Customers

CLUSTER	0	1	2	3
RECENCY	29.37	23.34	21.38	91.69
FREQUENCY	20.61	20.60	24.78	4.89
MONETARY	1154.98	784.49	997.93	167.18
AVG_BASKET_GROSS	10.76	10.86	11.16	13.20
CARD_USAGE_RATE	0.90	0.04	0.04	0.05
CATEGORY_DIVERSITY	8.95	8.13	8.04	5.72
TENURE_DAYS	3308.88	2099.63	5707.32	1556.30
AVG_PTS_EARNED	340.57	210.12	211.52	325.67
REDEMPTION_FREQ	0.06	0.04	0.04	0.02
DISCOUNT_SENSITIVITY	0.01	0.01	0.00	0.04
N_CUSTOMERS	6684.00	6959.00	7451.00	4634.00
PCT_%	26.00	27.00	29.00	18.00

- **Incredibly Loyal: These customers have been with the brand the longest, boasting an average tenure of over 15 years.**
- **Highly Active: They shop very frequently, averaging nearly 25 transactions over the measured period.**
- **Reliable Spenders: They bring in consistent revenue, averaging around €998 in total monetary value.**
- **Physical Preference: such type essentially prefer to buy in-store, which leads to more in-store interaction**
- **Strategy: retention based on in-store experience - digital promotional campaigns would be a waste of budget.**

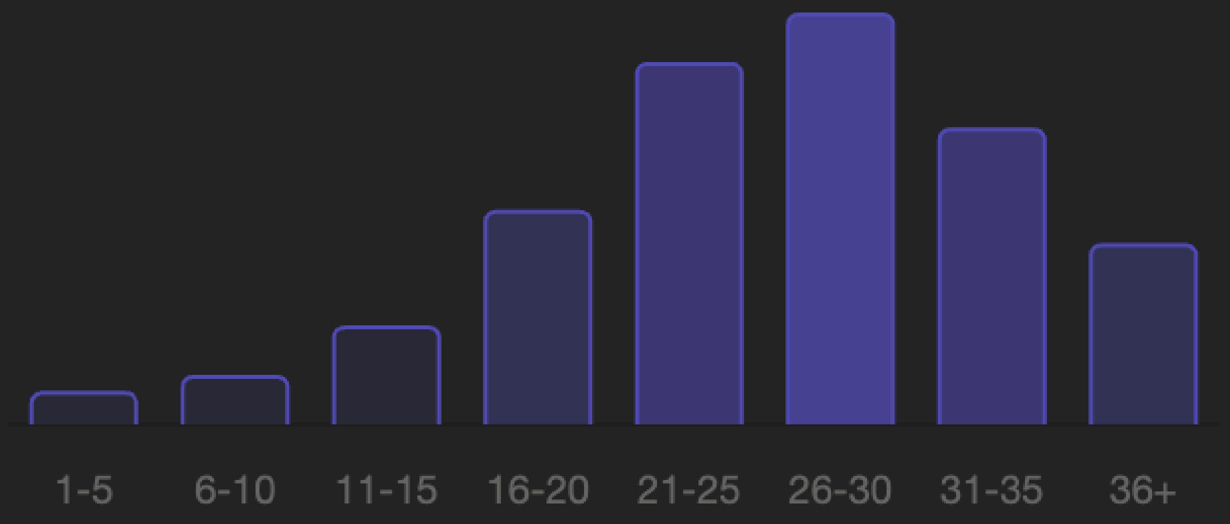
Historical Traditionalists (cluster 2)

29% of base · 7,451 customers · Loyal bedrock, cash-first, habit-driven

Visit frequency distribution

~25 visits

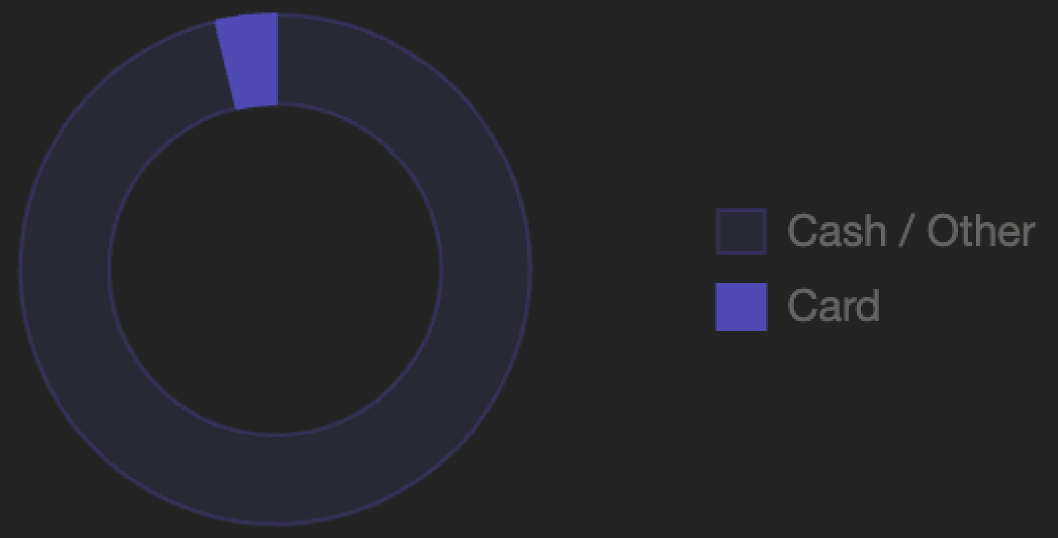
Highest frequency of all segments



Card usage rate

3.7%

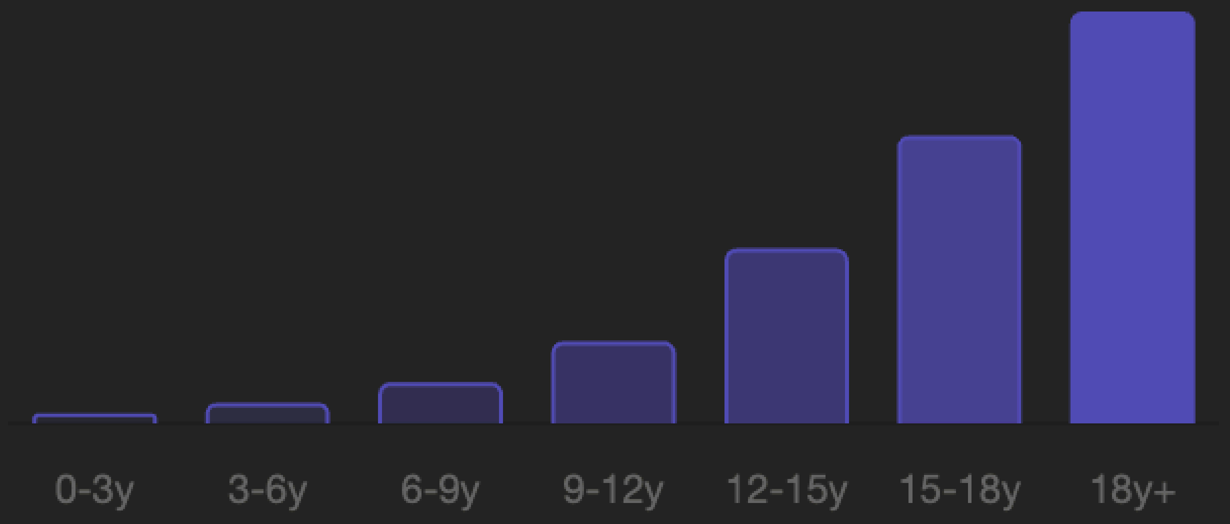
Near-zero — overwhelmingly cash buyers



Tenure distribution (years)

15.6 yrs avg

Deepest roots of any segment



Discount sensitivity vs spend

€998 avg

High spend, near-zero promo dependency

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Recent Regulars (cluster 1)

27% of base · 6,959 customers · Habit-formed, younger tenure, basket-building opportunity

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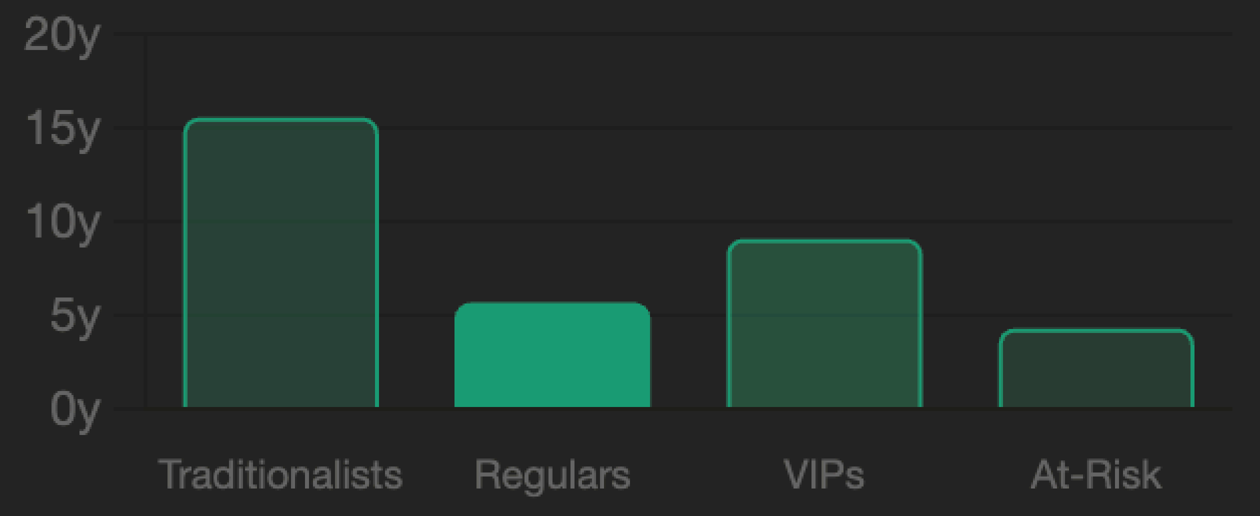
Tenure vs segment peers (years)

5.7 yrs avg

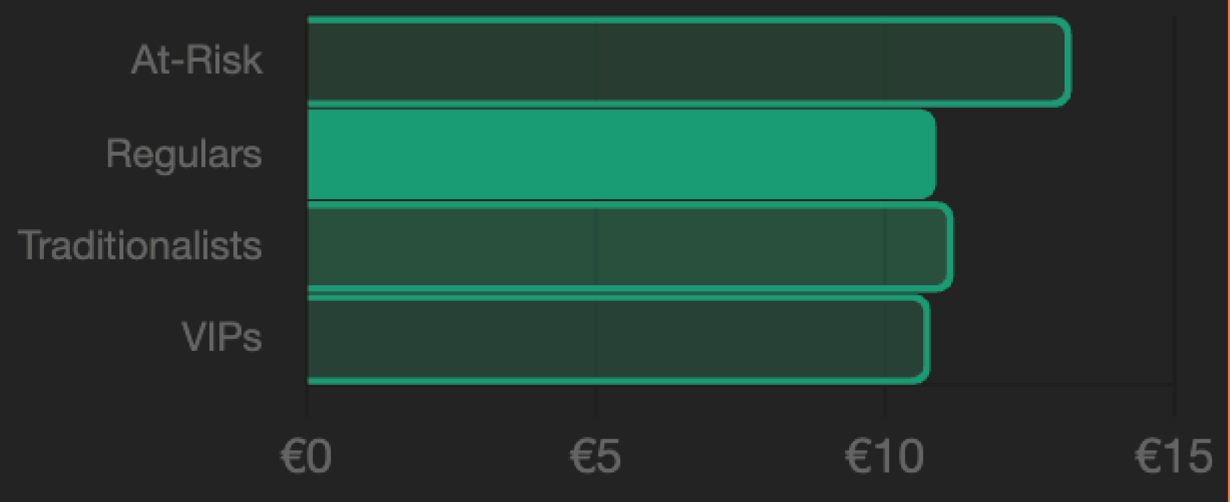
Youngest segment — still building loyalty



Average basket size

€10.86

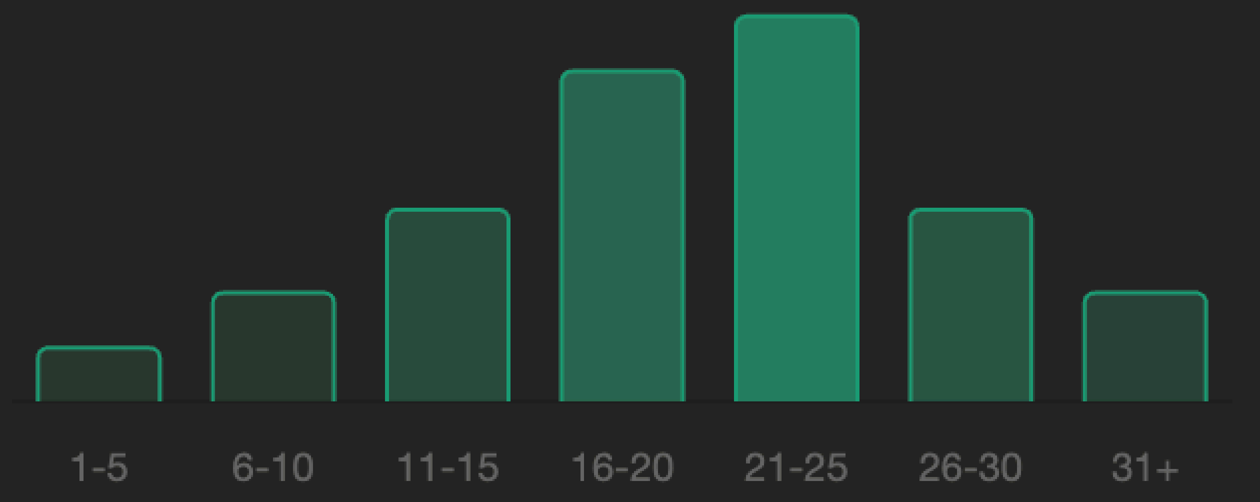
Lower basket vs Traditionalists (€11.16)



Visit frequency distribution

~21 visits

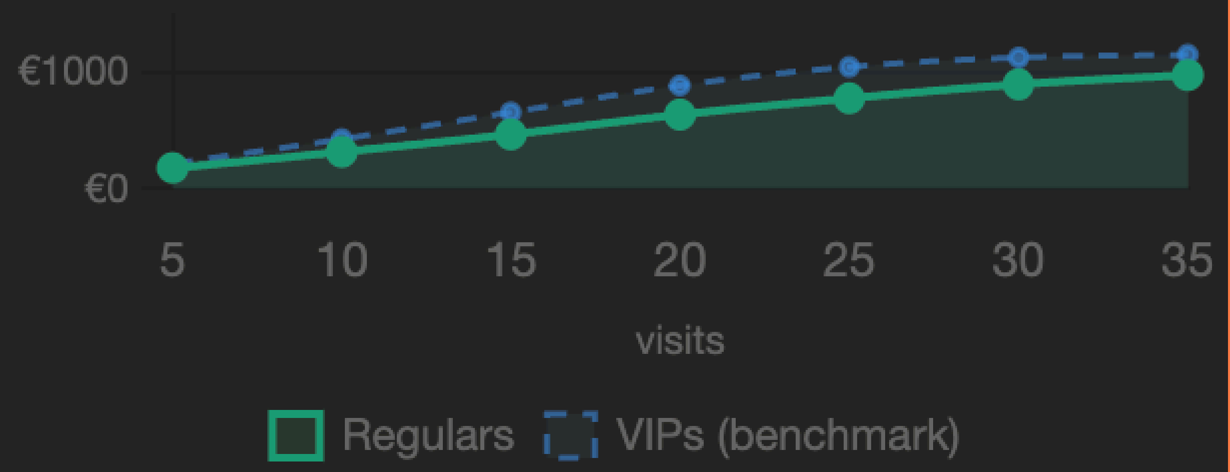
Solid habit already formed



Monetary spend vs frequency

€784 avg

Gap vs VIPs (€1,155) is the opportunity



- **The New Core: These are relatively newer shoppers, averaging about 5-6 years of tenure with the brand.**
- **Frequent Buyers: They have quickly adopted a strong habit, shopping just as frequently (about 21 times) as the VIPs.**
- **Solid Value: They spend a healthy amount, generating about €784 on average.**
- **Practical Choices: Like the others, "Ready-Made" goods are their most frequent purchases.**
- **Strategy: basket-building. Cross-selling campaigns to increase spend per visit and gradually transition them into higher-value tiers.**

- **Top Tier Spenders: This group generates the highest revenue, spending an average of €1,155 per customer.**
- **Strongly diversified: Throghout all 4 segments these type of customers have hihgly diversified category-preference**
- **Paid by card: Essential trait of such clientary base is to pay with card, cash/other payment methods used rarely**
- **Online First: As their name implies, they heavily favor interacting and purchasing through the Web.**
- **Strategy: The most important of engage with such clients is through app-based/ personalised marketing communication.**

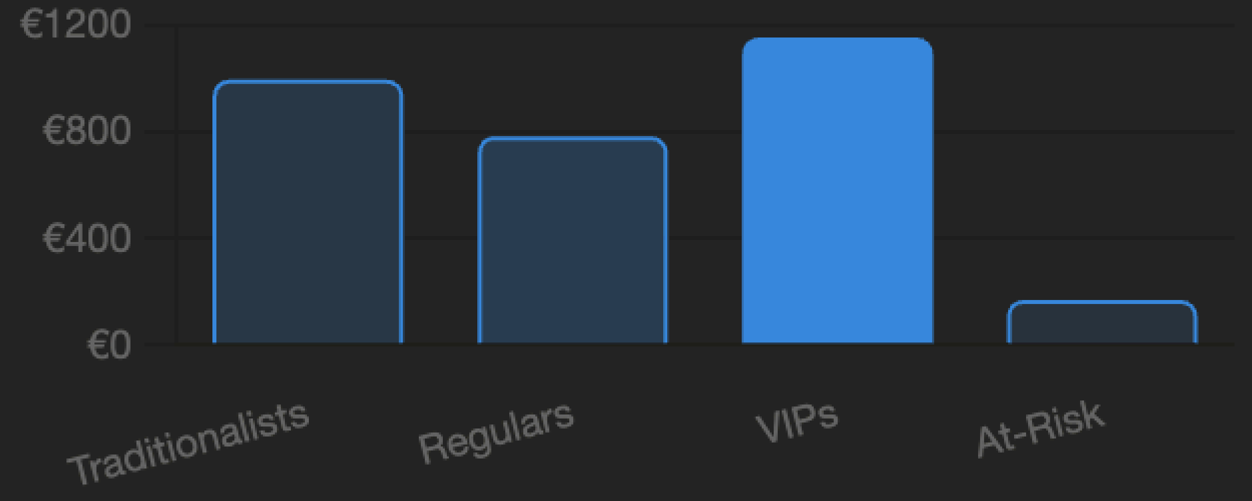
Digital VIPs (cluster 0)

26% of base · 6,684 customers · Economic engine — highest spend, fully digitized, category explorers

Monetary spend vs all segments (€)

€1,155 avg

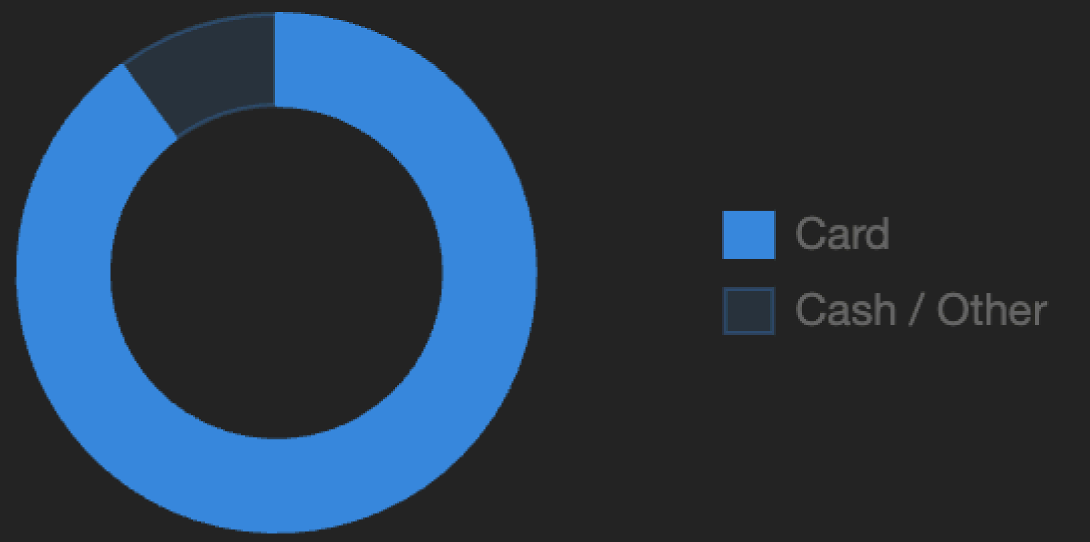
47% more than the next segment



Card usage rate

89.8%

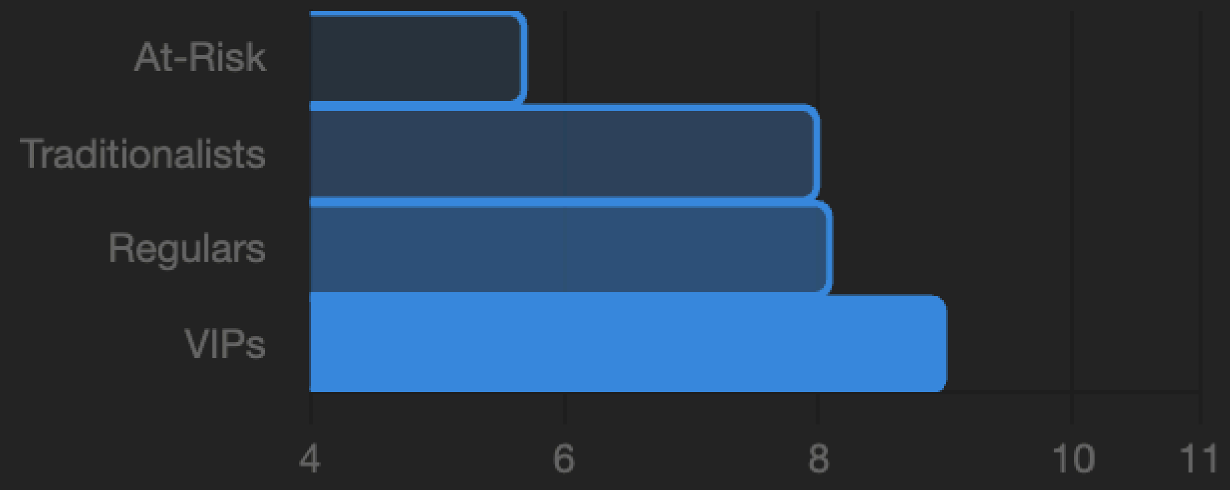
Overwhelmingly digital payment preference



Category diversity score

9.0 categories

Broadest explorers across all segments



Avg loyalty points earned

341 pts

Highest earner — active point redeemers

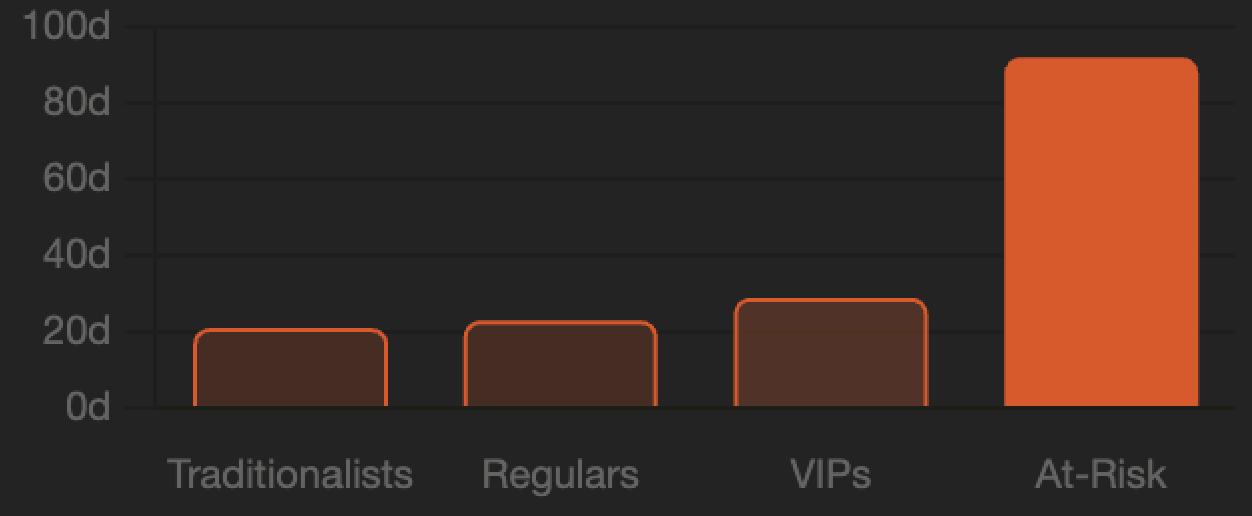
At-Risk / Promo-Driven (cluster 3)

18% of base · 4,634 customers · Severely disengaged — high recency, promo-only activation

Recency vs all segments (days ago)

92 days

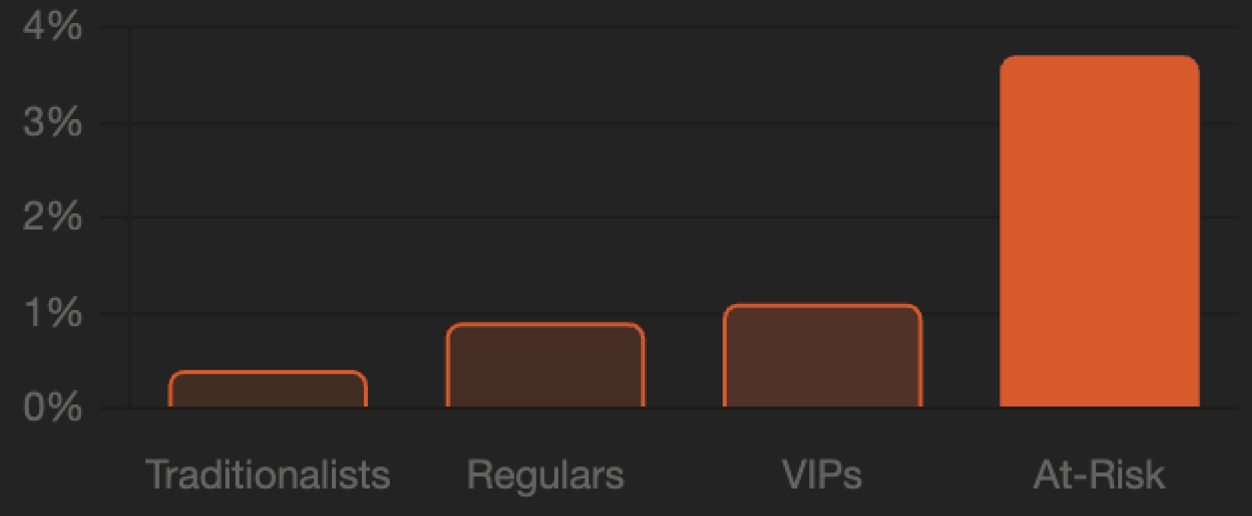
4× longer absent than any other segment



Discount sensitivity vs peers

3.7% rate

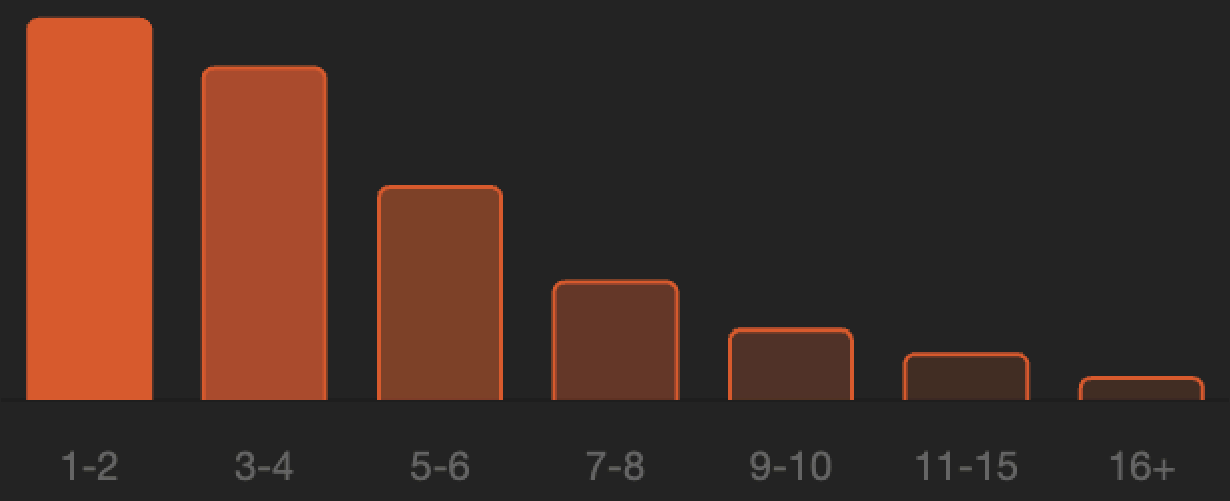
9× higher sensitivity than average



Visit frequency distribution

~5 visits

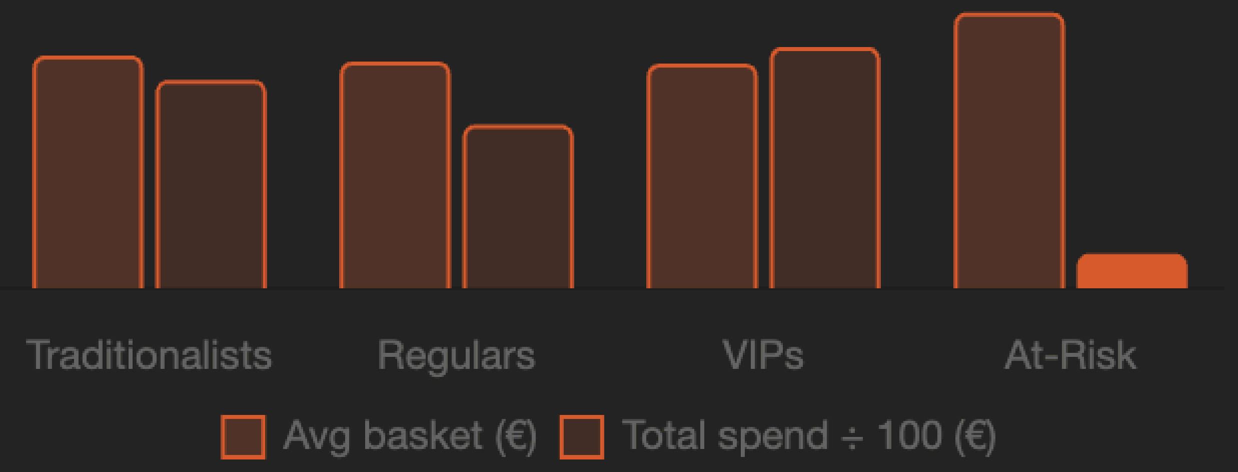
Lowest frequency — only promo-triggered



Spend vs basket size paradox

€167 total

High basket (€13.20) but very few trips



- **Fading Engagement: It has been an average of 3 months (over 90 days) since their last purchase.**
- **Low Frequency: They shop very rarely, completing fewer than 5 transactions on average.**
- **Low Spending: Their total monetary value is the lowest of all groups, sitting at just €167.**
- **Extreme Promo-Sensitivity: These customers are highly driven by discounts, exhibiting a 3.7% sensitivity rate nine times higher than the average shopper.**
- **Strategy: they need Re-engagement, as they are prime candidates for targeted discounts or win-back campaigns to prevent churn.**

DEPLOYMENT & CONCLUSION

READY FOR CRM

From Raw Data to Strategic Asset

Over 2.04 million transactions refined into 4 distinct, highly profitable personas.

100% Transparency

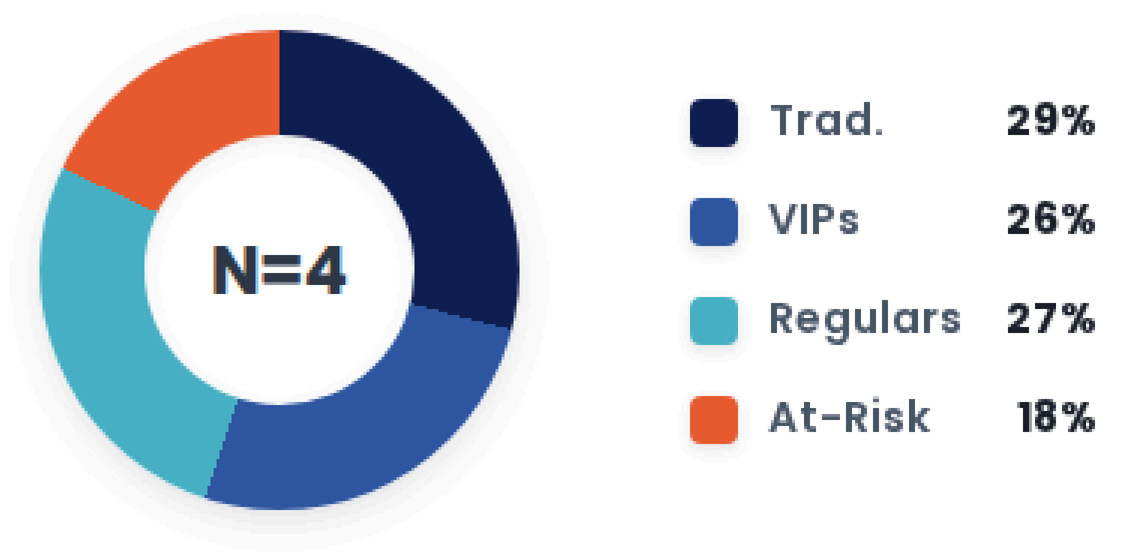
Every customer is mapped to tangible, original business metrics (No 'Black Box' algorithms).

Deployment Ready

Final clustering model successfully executed and complete mapping table generated.

Next Steps

Immediate ingestion into Var Group's campaign management platforms for localized and national Italian marketing.



~/var_group/output/customer_segments.csv

CUSTOMER_ID	CLUSTER	SEGMENT_NAME
CUST_008912	0	Digital VIP
CUST_114055	2	Historical Trad.
CUST_092334	1	Recent Regular
CUST_205661	3	At-Risk / Promo
[25,103 rows x 3 columns]		

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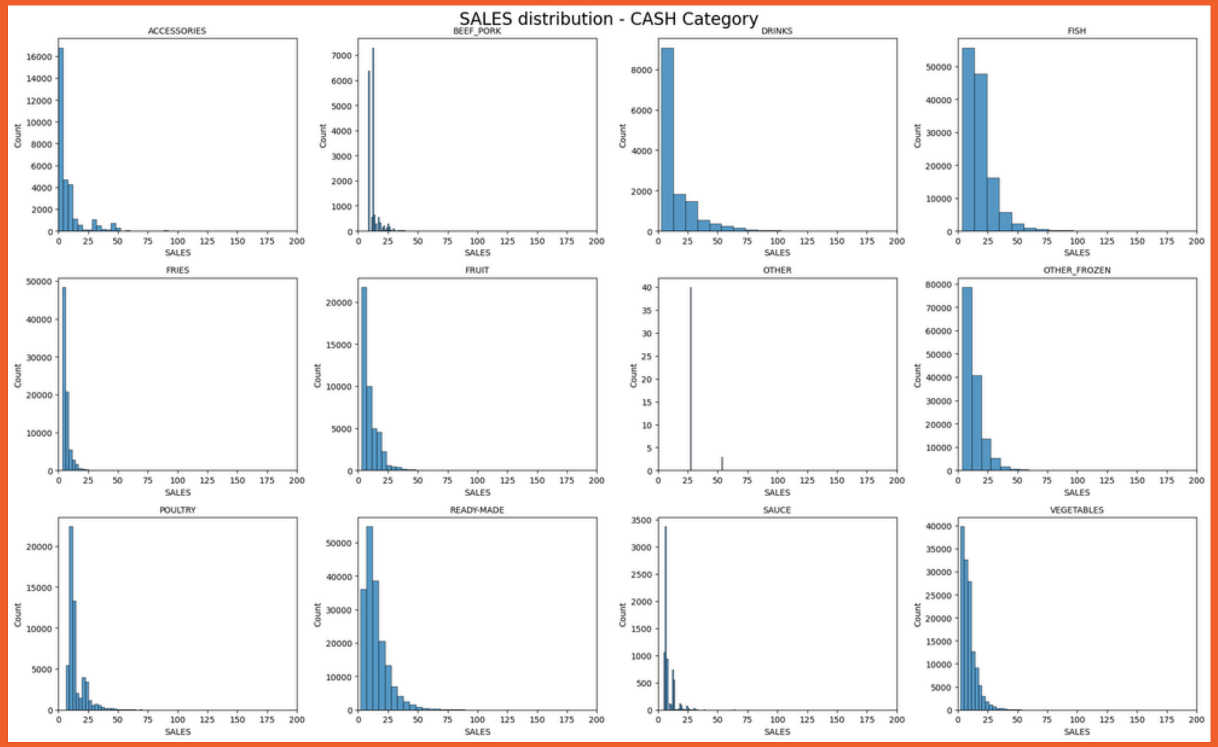
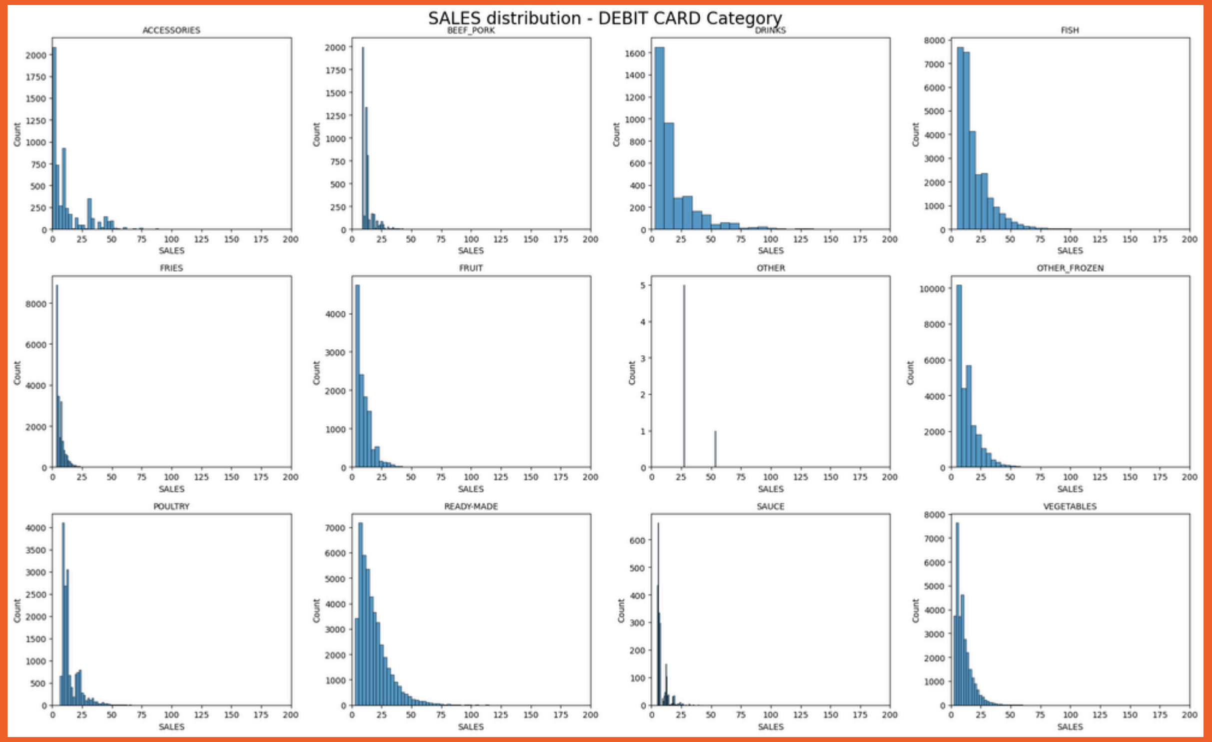
Appendix

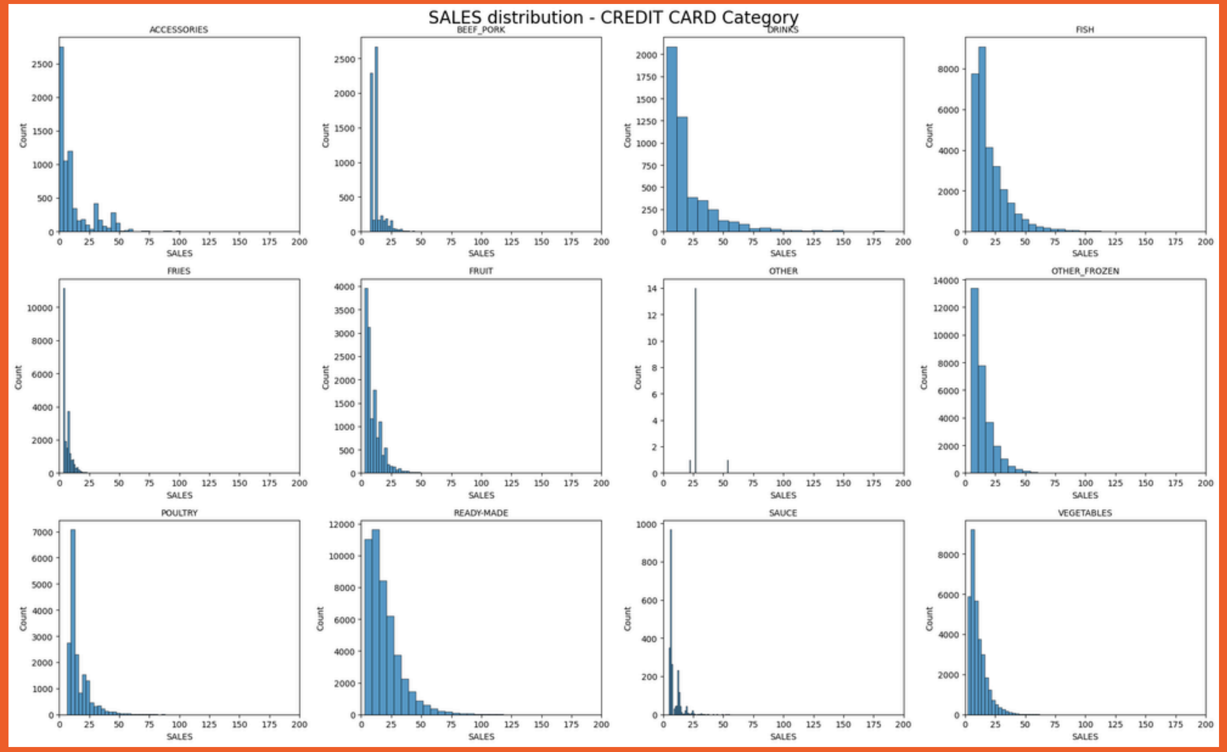
CASH



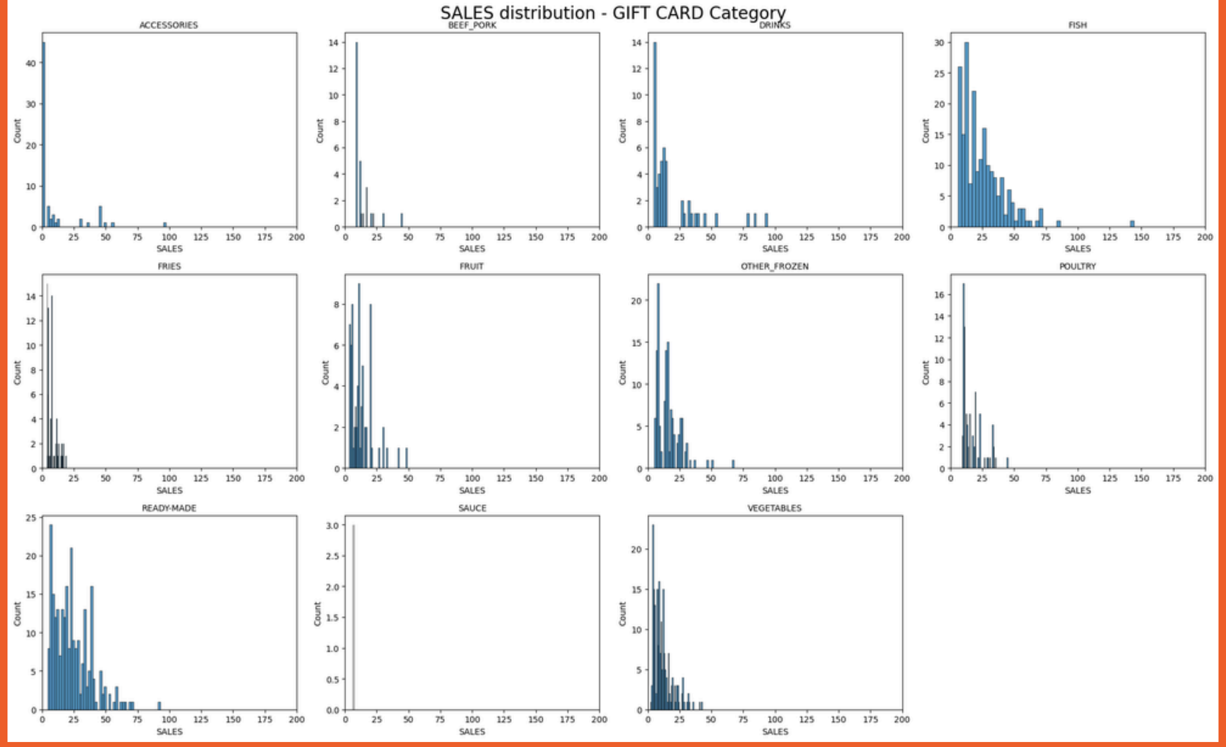
DEBIT CARD



CREDIT CARD

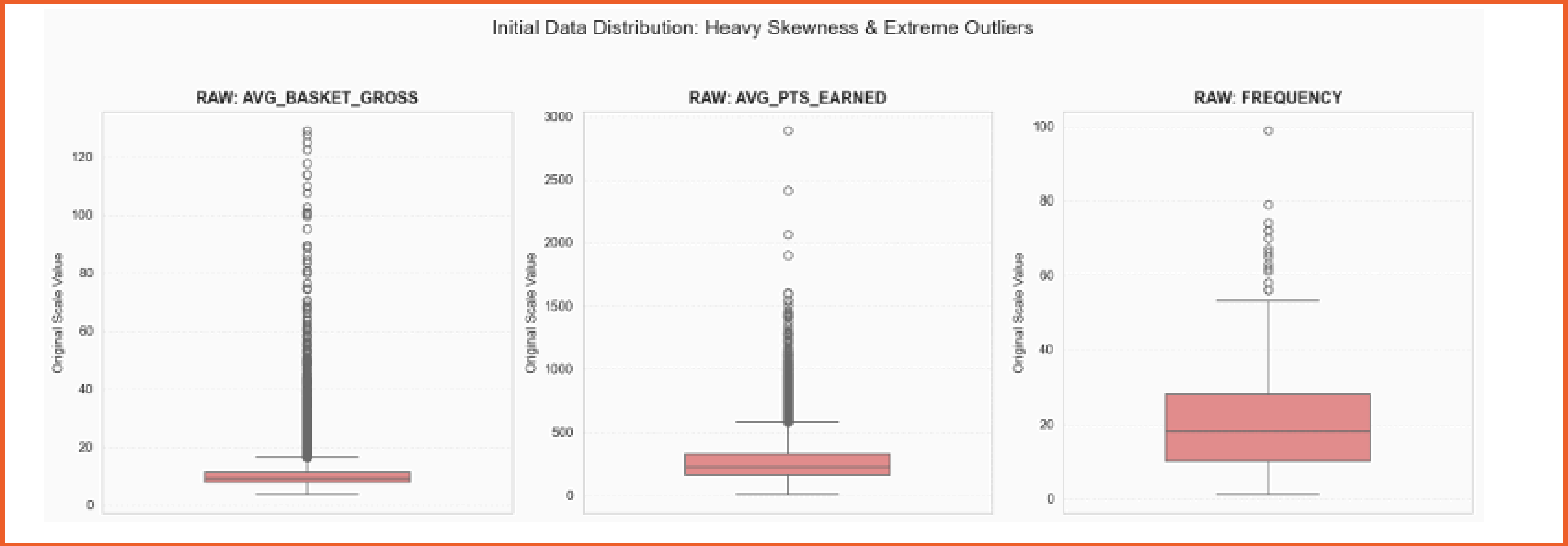


GIFT CARD



Appendix

BEFORE capping



AFTER capping